

1.7 EXCAVATION & BACKFILL LOG

Project Name:	Excavation Contractor:
Property Address:	Contact Phone:
Owner-Builder:	Equipment Used:

PURPOSE: This log documents all excavation and backfill activities for the foundation. Accurate records are essential for verifying code compliance, documenting soil conditions, and supporting warranty claims. Take photos at each stage.

SECTION 1: PRE-EXCAVATION INFORMATION

☐	Pre-Excavation Checklist	Date Completed	Notes
☐	811 utility locate completed - Ticket #: 		
☐	All utility locations marked and photographed		
☐	Building permit approved and posted on site		
☐	Foundation layout staked and squared		
☐	Erosion control measures in place		
☐	Tree protection fencing installed (if applicable)		

SECTION 2: EXCAVATION WORK LOG

Excavation Schedule

Excavation Start Date:		Start Time:	
Excavation Completion Date:		Completion Time:	
Total Excavation Duration:		Weather Conditions:	

Excavation Depth Measurements

Required Excavation Depth (per plan): _____ inches below finished grade

Local Frost Depth Requirement: _____ inches

Measurement Location	Required Depth (in)	Actual Depth (in)	Pass/Fail	Notes
Northeast (NE) Corner	_____	_____	☐ Pass ☐ Fail	_____
Northwest (NW) Corner	_____	_____	☐ Pass ☐ Fail	_____
Southeast (SE) Corner	_____	_____	☐ Pass ☐ Fail	_____
Southwest (SW) Corner	_____	_____	☐ Pass ☐ Fail	_____
North Wall Midpoint	_____	_____	☐ Pass ☐ Fail	_____
South Wall Midpoint	_____	_____	☐ Pass ☐ Fail	_____
East Wall Midpoint	_____	_____	☐ Pass ☐ Fail	_____
West Wall Midpoint	_____	_____	☐ Pass ☐ Fail	_____
Center of Excavation	_____	_____	☐ Pass ☐ Fail	_____
Other: _____	_____	_____	☐ Pass ☐ Fail	_____

SECTION 3: SOIL CONDITIONS DOCUMENTATION

Soil Type & Classification

Primary Soil Type:	☐ Clay ☐ Silt ☐ Sand ☐ Gravel ☐ Loam ☐ Mixed ☐ Other: _____
Soil Color:	_____
Soil Moisture Level:	☐ Dry ☐ Slightly Moist ☐ Wet ☐ Saturated ☐ Standing Water
Soil Bearing Capacity:	_____ PSF (per geotechnical report or code assumption)
Organic Matter Present:	☐ None ☐ Minimal ☐ Moderate ☐ Significant (requires removal)
Frost-Susceptible Soil:	☐ Yes ☐ No - Action Taken: _____

Rock, Ledge & Obstacles Encountered

Location	Depth Found (in)	Type (rock, ledge, debris)	Action Taken	Notes
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____

Water & Drainage Issues

Standing Water Encountered:	☐ Yes ☐ No - Location: _____
Groundwater Seepage:	☐ Yes ☐ No - Depth: _____ inches Flow Rate: ☐ Slow ☐ Moderate ☐ Fast
Dewatering Required:	☐ Yes ☐ No - Method: ☐ Pump ☐ Sump ☐ Gravity Drain ☐ Other: _____
High Water Table Noted:	☐ Yes ☐ No - Depth from surface: _____ feet

Drainage Improvements Made:	
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Excavated Material Disposition

Material Type	Estimated Volume	Disposition	Location/Notes
Topsoil (saved for landscaping)	_____ cu. yds	☐ Stockpiled ☐ Removed	
Subsoil (reusable for backfill)	_____ cu. yds	☐ Stockpiled ☐ Removed	
Rock/gravel	_____ cu. yds	☐ Stockpiled ☐ Removed	
Unsuitable material (organic, wet)	_____ cu. yds	☐ Removed ☐ Hauled away	

SECTION 4: BACKFILL OPERATIONS LOG

Backfill Schedule

Backfill Start Date:	_____	Days Since Concrete Pour:	_____ days
Backfill Completion Date:	_____	Weather Conditions:	_____
Waterproofing Cured:	☐ Yes ☐ No ☐ N/A	Drainage Tile Installed:	☐ Yes ☐ No ☐ N/A

Backfill Material Specifications

Primary Backfill Material:	☐ Excavated Soil ☐ Clean Fill ☐ Sand ☐ Gravel ☐ Crushed Stone ☐ Other: _____
Material Source:	☐ On-site ☐ Imported - Supplier: _____
Material Quality:	☐ Free of organic matter ☐ Free of large rocks (>3") ☐ Properly graded
Moisture Content:	☐ Optimum for compaction ☐ Too wet ☐ Too dry - Action: _____
Total Backfill Volume:	_____ cubic yards

Compaction Method & Testing

Compaction Equipment Used:	☐ Plate Compactor ☐ Jumping Jack ☐ Roller ☐ Hand Tamper ☐ Other: _____
Lift Thickness:	_____ inches (typically 6-12 inches before compaction)
Number of Passes per Lift:	_____ passes with compactor
Total Number of Lifts:	_____ lifts
Compaction Testing Performed:	☐ Yes ☐ No - Method: ☐ Nuclear Density ☐ Proctor ☐ Visual ☐ Other: _____
Target Compaction Level:	_____ % of maximum dry density (typically 90-95% for backfill)

Compaction Test Results

Test Location	Lift Depth (in)	Compaction % Achieved	Pass/Fail	Notes/Actions
NE Corner	_____	_____ %		_____

Test Location	Lift Depth (in)	Compaction % Achieved	Pass/Fail	Notes/Actions
			☐ Pass ☐ Fail	
NW Corner	_____	_____ %	☐ Pass ☐ Fail	_____
SE Corner	_____	_____ %	☐ Pass ☐ Fail	_____
SW Corner	_____	_____ %	☐ Pass ☐ Fail	_____
North Wall	_____	_____ %	☐ Pass ☐ Fail	_____
South Wall	_____	_____ %	☐ Pass ☐ Fail	_____
East Wall	_____	_____ %	☐ Pass ☐ Fail	_____
West Wall	_____	_____ %	☐ Pass ☐ Fail	_____

SECTION 5: FINAL GRADING & DRAINAGE

☐	Final Grading Checklist	Date Completed	Notes
☐	Backfill completed to within 6-8" of final grade	_____	_____
☐	Grade slopes away from foundation (2% min, 6" drop in 10 feet)	_____	_____
☐	No low spots or depressions where water can collect	_____	_____
☐	Swales or drainage paths direct water away from structure	_____	_____
☐	Perimeter drain outlet is clear and functional	_____	_____
☐	Final topsoil layer applied (if applicable)	_____	_____
☐	Erosion control measures in place until landscaping	_____	_____
☐	Grading verified after first significant rain event	_____	_____

SECTION 6: PHOTO DOCUMENTATION

Photographic evidence is critical for insurance claims, warranty issues, and inspection appeals. Take multiple photos from different angles at each stage.

Date	Photo Description	File Name/Number	Notes
_____	Site before excavation begins	_____	_____
_____	Utility locate markings	_____	_____
_____	Foundation layout stakes/strings	_____	_____
_____	Excavation in progress	_____	_____
_____	Final excavation depth verification	_____	_____
_____	Soil conditions/layers exposed	_____	_____
_____	Rock or unusual conditions found	_____	_____
_____	Water/drainage issues encountered	_____	_____
_____		_____	_____

Date	Photo Description	File Name/Number	Notes
	Foundation complete before backfill		
	Waterproofing applied		
	Drainage tile installation		
	Backfill in progress (show lifts)		
	Compaction equipment in use		
	Final backfill completed		
	Final grading and drainage		

SECTION 7: CONTRACTOR SIGN-OFF & CERTIFICATION

I certify that all excavation and backfill work has been completed in accordance with approved plans and industry best practices. All compaction meets or exceeds specified requirements.

Excavation Contractor Signature:

Contractor Name (printed):

Company:

License # (if applicable):

Date:

Owner-Builder Acceptance:

Date:

NOTES, ISSUES & LESSONS LEARNED

